Experiment No.:5

Name of Experiment: Use R as a calculator.

Aim: To make a calculator in R.

Source Code: Basic Calculations in R

1. Basic Arithmetic Operations

```
#Addition
add_result <- 15 + 7

#Subtraction
sub_result <- 20 - 8

#Multiplication
mul_result <- 6 * 9

#Division
div_result <- 81 / 9

#Exponentiation (raising to the power)
exp_result <- 5^3   #5 raised to the power of 3

#Modulus (remainder of division)
mod_result <- 17 %% 5

#Integer division (quotient without remainder)
int_div_result <- 17 %/% 5

#Square root
sqrt_result <- sqrt(144)

#Logarithm (log base e)
log_result <- log(2.718)

#Trigonometric functions (sin, cos, tan)
sin_result <- sin(pi / 2)   #sin of 90 degrees (in radians)
cos_result <- cos(0)        #cos of 0 degrees (in radians)
tan_result <- tan(pi / 4)   #tan of 45 degrees (in radians)

#Print the results
print(paste("Addition: 15 + 7 =", add_result))
print(paste("Subtraction: 20 - 8 =", sub_result))
print(paste("Multiplication: 6 * 9 =", mul_result))
print(paste("Division: 81 / 9 =", div_result))
print(paste("Exponentiation: 5^3 =", exp_result))
print(paste("Modulus: 17 %% 5 =", mod_result))
print(paste("Integer Division: 17 %/% 5 =", int_div_result))
print(paste("Square Root: sqrt(144) =", sqrt_result))
print(paste("Logarithm of e: log(2.718) =", log_result))
print(paste("Sine of 90 degrees: sin(pi/2) =", sin_result))
print(paste("Cosine of 0 degrees: cos(0) =", cos_result))
print(paste("Tangent of 45 degrees: tan(pi/4) =", tan_result))
```

Output:

```
 [1] "Addition: 15 + 7 = 22"
[1] "Subtraction: 20 - 8 = 12"
[1] "Multiplication: 6 * 9 = 54"
[1] "Division: 81 / 9 = 9"
[1] "Exponentiation: 5^3 = 125"
[1] "Modulus: 17 %% 5 = 2"
[1] "Integer Division: 17 %/% 5 = 3"
[1] "Square Root: sqrt(144) = 12"
[1] "Logarithm of e: log(2.718) = 1.000422"
[1] "Sine of 90 degrees: sin(pi/2) = 1"
[1] "Cosine of 0 degrees: cos(0) = 1"
[1] "Tangent of 45 degrees: tan(pi/4) = 1"
```